

Ting Chun Liu

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EDUCATION

Johns Hopkins University, Baltimore, MD

Aug 2026 – Expected May 2028

M.S.E in Biomedical Engineering, Imaging & Medical Devices Track

| Current Coursework: Computer Integrated Surgery, Sensor-Based Robotics, Probabilistic Machine Learning

National Taiwan University, Taipei, Taiwan

Sep 2021 - Jun 2025

B.S in Electrical Engineering, Overall GPA: 3.81/4.0

| Relevant Coursework: Signals and Systems, Linear Algebra, Probability and Statistics, Data Structures, Algorithms, Machine Learning, Computer Vision, Reinforcement Learning, Security and Privacy of ML

RESEARCH

Machine Learning for Meaningful Lives Lab, NTU – *Undergraduate Researcher*

Jul 2025 - Sep 2025

Supervisor: Prof. Tzu-Yu Liu

Deep Learning-Based Diagnostic System for Colonoscopy Videos / Team size: 5

- ▶ Developed PyTorch-based models to segment and classify colorectal lesions using clinical patient data.
- ▶ Applied transfer learning and explainable AI techniques, including Integrated Gradients, to improve model performance and interpretability under limited clinical data.
- ▶ Built a privacy-aware video labeling platform for physicians to enable efficient and robust data collection.

MAGICS Lab, Northwestern University – *Research Assistant (Remote)*

Oct 2024 - Apr 2025

Supervisor: Prof. Han Liu

Collocation-Based ODE Solvers for Score-Based Generative Models / Team size: 6

- ▶ Designed and conducted synthetic and real-world experiments to validate collocation-based ODE solvers for score-based generative models; benchmarked Fourier, Lagrange, and RBF basis functions across multiple configurations.
- ▶ Identified computational bottlenecks on CelebA-HQ and LSUN Church arising from basis-function assumptions, informing the method's scalability analysis.

INTERNSHIP

CancerFree BioTech, Taipei, Taiwan – *Engineering Intern, Software & Product Development*

Sep 2024 - Feb 2026

Smart Glasses for Trustworthy and Hands-Free Wet-Lab Documentation

- ▶ Fine-tuned 3B/7B vision-language models (Qwen2.5-VL-Instruct, InternVideo2.5) on customized QA pairs for dense video captioning using PyTorch.
- ▶ Developed an end-to-end visual data pipeline for video ingestion, preprocessing, semantic retrieval, model post-training, and robust report generation using Python, NVIDIA VSS, Supabase, and Docker.

Deep Learning for Robust Cell Image Analysis

- ▶ Fine-tuned Cellpose Foundation model on
 1. fluorescence-stained images to extract cell-level statistics and quantify circulating tumor cells (CTCs).
 2. brightfield images to segment target cells and inform imaging-device parameter adjustments.
- ▶ Deployed the complete system using Docker to enable quick and consistent access for lab members.

Ensemble AI, San Francisco, CA, USA – *AI Researcher Intern (Remote)*

Feb 2025 - May 2025

Structure-Aware Tensor Decomposition for Parameter-Efficient Neural Networks [[arXiv:2503.17353](https://arxiv.org/abs/2503.17353)]

- ▶ Implemented drop-in NdLinear layers for Transformers / CNNs / RNNs / task-specific architectures to replace standard linear layers, reducing parameters by 50–95% for tabular / time-series / vision tasks with consistent accuracy improvements.
- ▶ Pre-trained OPT-124M and OPT-350M models on BookCorpus and Wiki40B-English, achieving lower perplexity under identical compute budgets, confirming the scaling benefits of NdLinear.